

SKILL WEEK 6

Task 1:

Code:

```
#include <stdio.h>

int main() {
    char input[200];
    int i;

    printf("Enter a string with escape characters: ");
    fgets(input, sizeof(input), stdin);

    printf("\nProcessed string:\n");
    for (i = 0; input[i] != '\0'; i++)
    {
        if (input[i] == '\\' && input[i + 1] !=
            '\0') {
            i++;
            switch (input[i])
            {
                case
                'n':
                    printf("\n");
                    break;
                case 't':
                    printf("\t");
                    break;
                case ' ':
                    printf(" ");
                    break;
                case '\\':
                    printf("\\");
                    break;
                case '$':
                    printf("$");
                    break;
                case '!':
                    printf("!");
                    break;
                default:
                    printf("%c", input[i]);
            }
        }
    }
}
```

```

    } else {          printf("%c",
input[i]);
    }
}

return 0;
}

```

```

naveen123@LAPTOP-4NVQJ2C7:~$ gcc skill6.c -o skill6
naveen123@LAPTOP-4NVQJ2C7:~$ ./skill6
Enter a string with escape characters: Hello\nWorld

Processed string:
Hello
World
naveen123@LAPTOP-4NVQJ2C7:~$ |

```

Task 2:

Code:

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()

{ pid_t pid;

    pid = fork();
    if (pid < 0) {
        perror("fork failed");    return
        1;
    }

    if (pid == 0) {

        // Child process    printf("Child
        process created.\n");    printf("Child
        PID: %d\n", getpid());

        char *args[] = {"ls", "-l", NULL};    execvp(args[0], args);

        // This runs only if execvp fails
        perror("execvp failed");    exit(1);
    }
}

```

```
    } else {  
  
        // Parent process    printf("Parent  
process.\n");    printf("Parent PID:  
%d\n", getpid());  
  
        wait(NULL);  
  
        printf("Child process finished.\n");  
    }  
  
    return 0;  
}
```

Output:

```

naveen123@LAPTOP-4NVQJ2C7:~$ skill6.1.c
skill6.1.c: command not found
naveen123@LAPTOP-4NVQJ2C7:~$ nano skill6.1.c
naveen123@LAPTOP-4NVQJ2C7:~$ gcc skill6.1.c -o skill6.1
naveen123@LAPTOP-4NVQJ2C7:~$ ./skill6.1
Parent process.
Parent PID: 1087
Child process created.
Child PID: 1088
total 31468
-rw-r--r--  1 naveen123 naveen123 10485760 Sep 29 03:14 Naveen.txt
drwxr-xr-x 13 naveen123 naveen123   4096 Aug 20 06:09 OSSP
-rwxr-xr-x  1 naveen123 naveen123  16376 Aug 31 05:22 a.out
-rwxr-xr-x  1 naveen123 naveen123  16384 Aug 31 05:12 client
-rw-r--r--  1 naveen123 naveen123    558 Aug 31 05:12 client.c
-rwxr-xr-x  1 naveen123 naveen123  16400 Aug  3 05:32 command_executor
-rw-r--r--  1 naveen123 naveen123   1116 Aug  3 05:31 command_executor.c
-rwxr-xr-x  1 naveen123 naveen123  16392 Aug 24 04:31 exp1
-rw-r--r--  1 naveen123 naveen123    649 Aug 24 04:31 exp1.c
-rwxr-xr-x  1 naveen123 naveen123  16296 Aug 24 04:16 exp2
-rw-r--r--  1 naveen123 naveen123   1065 Aug 24 04:16 exp2.c
-rwxr-xr-x  1 naveen123 naveen123  16208 Aug 24 05:27 exp3
-rw-r--r--  1 naveen123 naveen123    598 Aug 24 05:27 exp3.c
-rwxr-xr-x  1 naveen123 naveen123  16288 Aug 24 05:32 exp4
-rw-r--r--  1 naveen123 naveen123    509 Aug 24 05:32 exp4.c
-rwxr-xr-x  1 naveen123 naveen123  16248 Sep 30 06:49 exp7
-rw-r--r--  1 naveen123 naveen123    846 Sep 30 06:49 exp7.c
-rwxr-xr-x  1 naveen123 naveen123  16296 Sep 30 06:55 exp71
-rw-r--r--  1 naveen123 naveen123    730 Sep 30 06:55 exp71.c
-rwxr-xr-x  1 naveen123 naveen123  16168 Sep 30 07:02 exp8
-rw-r--r--  1 naveen123 naveen123   1201 Sep 30 07:02 exp8.c
-rwxr-xr-x  1 naveen123 naveen123  16288 Sep 30 07:04 exp81
-rw-r--r--  1 naveen123 naveen123   1424 Sep 30 07:03 exp81.c
-rwxr-xr-x  1 naveen123 naveen123  16384 Aug 18 04:14 fork

```